MIDTERM - I

1.

$\boxed{n=2}$

$$\psi_2(x) = \sqrt{\frac{2}{a}} \sin\left[\frac{2\pi}{a} x\right]$$

$$P_{(2a/3,\, a)} = \int_{2a/3}^{a} \psi_2^*(x)\, \psi_2(x)\, dx$$

$$= \left(\frac{2}{a}\right) \int_{2a/3}^{a} \sin^2\left[\frac{2\pi}{a} x\right] dx$$

$$\cos 2\theta = 1 - 2\sin^2\theta \quad \Rightarrow \quad \sin^2\theta = \frac{1-\cos 2\theta}{2}$$

$$\Rightarrow P_{(2a/3,\, a)} = \frac{2}{a} \int_{2a/3}^{a} \frac{(1-\cos 2\theta)}{2}\, dx \quad ; \; 2\theta = \frac{4\pi}{a} x$$

$$= \frac{2}{a}\left[\frac{x}{2} - \frac{1}{2}\, \frac{\sin\left(\frac{4\pi}{a} x\right)}{(4\pi/a)}\right]_{2a/3}^{a}$$

$$= \frac{2}{a}\left[\frac{a}{6} + \frac{a}{2}\, \frac{\sin\left(\frac{8\pi}{3}\right)}{4\pi}\right]$$

$$= \frac{1}{3} + \frac{\sqrt{3}}{8\pi}$$

2.

$$\psi(x,0) = \frac{1+i}{2}\psi_0(x) + \frac{1}{\sqrt{2}}\psi_1(x)$$

(a) $$\psi(x,t) = \frac{1+i}{2}\psi_0 e^{-iE_0 t/\hbar} + \frac{1}{\sqrt{2}}\psi_1 e^{-iE_1 t/\hbar}$$

where $E_n = \hbar\omega\left(n+\frac{1}{2}\right)$

(b) Probability →

$$\frac{\hbar\omega}{2} \rightarrow P(E_0) = \left|\frac{1+i}{2}\right|^2 = \frac{1}{2}$$

$$\frac{3\hbar\omega}{2} \rightarrow P(E_1) = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$$

$$\frac{7\hbar\omega}{2} \rightarrow P(E_2) = 0$$

(c) $$\langle x\rangle = \int \psi^*(x,t)\left[\sqrt{\frac{\hbar}{2m\omega}}\,(a_+ + a_-)\right]\psi(x,t)$$

$$= \int \psi^*(x,t)\left[\sqrt{\frac{\hbar}{2m\omega}}\,(a_+\psi + a_-\psi)\right]dx$$

$$= \int \psi^*(x,t)\left[\sqrt{\frac{\hbar}{2m\omega}}\left[\left(\frac{1+i}{2}\right)\psi_1 e^{-iE_0 t/\hbar} + \frac{1}{\sqrt{2}}\sqrt{2}\,\psi_2 e^{-iE_1 t/\hbar}\right] + \sqrt{\frac{\hbar}{2m\omega}}\left[0 + \frac{1}{\sqrt{2}}\psi_{II}\, e^{-iE_1 t/\hbar}\right]\right]dx$$

use $\rightarrow$ $\displaystyle\int_{-\infty}^{\infty} \psi_m^* \psi_n \, dx = \delta_{mn}$

and $\psi^*(x,t) = \left(\frac{1-i}{2}\right) \psi_0^* e^{iE_0 t/\hbar} + \frac{1}{\sqrt{2}} \psi_1 e^{iE_1 t/\hbar}$

thus $\rightarrow$ $\langle x \rangle = \sqrt{\frac{\hbar}{2m\omega}} \left(\frac{1-i}{2}\right) \frac{1}{\sqrt{2}} e^{i(E_0 - E_1)t/\hbar}$

$$+ \sqrt{\frac{\hbar}{2m\omega}} \left(\frac{1+i}{2}\right) \frac{1}{\sqrt{2}} e^{i(E_1 - E_0)t/\hbar}$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \, 2 \, \mathrm{Re} \left[\frac{(1+i)}{2\sqrt{2}} e^{i(E_1 - E_0)t/\hbar} \right]$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \, \cancel{2} \left[\frac{\cos(E_1 - E_0)t/\hbar - \sin(E_1 - E_0)t/\hbar}{\cancel{2}\sqrt{2}} \right]$$

$$= \sqrt{\frac{\hbar}{4m\omega}} \left[\cos \omega t - \sin \omega t \right]$$

Similarly: $\langle p \rangle = \displaystyle\int \psi^*(x,t) \; i \sqrt{\frac{\hbar m\omega}{2}} (a_+ - a_-) \, \psi(x,t) \, dx$

$$= \int \psi^*(x,t) \, i \sqrt{\frac{\hbar m\omega}{2}} \left[\left(\frac{1+i}{2}\right) \psi_1 e^{-iE_0 t/\hbar} + \frac{\sqrt{2}}{\sqrt{2}} \psi_2 e^{-iE_1 t/\hbar} \right.$$

$$\left. - \left(0 + \frac{\psi_0}{\sqrt{2}} e^{-iE_1 t/\hbar} \right) \right]$$

After similar steps one obtains →

$$\langle p \rangle = -\sqrt{\frac{\hbar m \omega}{4}}\left[\cos\omega t + \sin\omega t\right]$$

(d) Ehrenfest's theorem :

$$\frac{d\langle p\rangle}{dt} = \left\langle -\frac{\partial V}{\partial x}\right\rangle$$

i.e to show that

$$\frac{d\langle p\rangle}{dt} = \left\langle -\frac{\partial}{\partial x}\left[\frac{1}{2}m\omega^2 x^2\right]\right\rangle$$

$$= -\frac{1}{2}m\omega^2\langle x\rangle$$

$$\frac{d\langle p\rangle}{dt} = -\sqrt{\frac{\hbar m\omega}{4}}\left[-\omega\sin\omega t + \omega\cos\omega t\right]$$

$$= -m\omega^2\left(\sqrt{\frac{\hbar}{4m\omega}}\left[\cos\omega t - \sin\omega t\right]\right)$$

$$= -m\omega^2\langle x\rangle$$

↳ Proved

3. $\psi(x) = A e^{-a|x|} \longrightarrow$ symmetric about $x=0$

(a) Normalize $\rightarrow$

$$\int_{-\infty}^{\infty} |\psi|^2 dx = 2\int_0^{\infty} A^2 e^{-2ax} dx = 1$$

$$\Rightarrow \frac{A^2}{a}\int_0^{\infty} e^{-2ax}\, d(2ax)$$

$$= \frac{A^2}{a}\, 1 = 1$$

$$\Rightarrow A = \sqrt{a}.$$

(b) $P_{(0,\infty)} = \frac{1}{2}$ $\leftarrow$ Due to symmetry

(c) $$P_{(1/a,\infty)} = \int_{1/a}^{\infty} \psi^* \psi\, dx = a\int_{1/a}^{\infty} e^{-2ax} dx$$

$$= -\frac{a}{2a}\left[e^{-2a}\right]_{1/a}^{\infty} = \frac{e^{-2}}{2}$$

(d) $$P_{(-1/a,\,1/a)} = 2\, P_{(0,\,1/a)} = 2\left[P_{(0,\infty)} - P_{(\frac{1}{a},\infty)}\right]$$

$$= 2\left[\frac{1}{2} - \frac{e^{-2}}{2}\right] = 1 - e^{-2}$$

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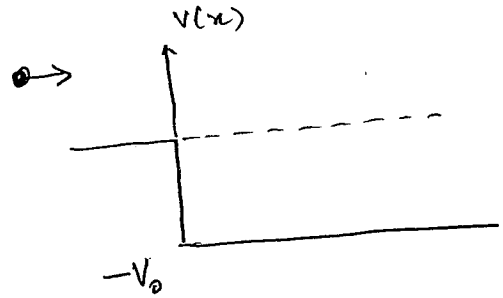

$E > 0$

Wave functions on the left hand side →

$$a e^{ikx} + r e^{-ikx}$$

where $k = \sqrt{\frac{2mE}{\hbar^2}} = \sqrt{\frac{2mV_0}{3\hbar^2}}$

Wave function on the right hand side →

$$t e^{ilx}$$

where $l = \sqrt{\frac{2m(E+V_0)}{\hbar^2}} = \sqrt{\frac{8mV_0}{3\hbar^2}}$

From continuity of wave function at $x = 0$

$$a e^{ikx} + r e^{-ikx}\Big|_{x=0} = t e^{ilx}\Big|_{x=0}$$

$$\Rightarrow \boxed{a + r = t} \longrightarrow (a)$$

From differentiability of wave function at $x = 0$

$$ik\left(a e^{ikx} - r e^{-ikx}\right)\Big|_{x=0} = ilt e^{ilx}\Big|_{x=0}$$

$$\Rightarrow \quad ik(a - r) = i l t$$

$$\Rightarrow \quad \boxed{(a - r) = \frac{l t}{k}} \longrightarrow (b)$$

subtracting (a) with (b)

$$(a + r) - (a - r) = t - \frac{l t}{k}$$

$$2r = t\left(1 - \frac{l}{k}\right)$$

or $$t = \frac{2r}{1 - \frac{l}{k}}$$

From (a) $\rightarrow$

$$a + r = \frac{2r}{1 - \frac{l}{k}}$$

$$\Rightarrow \quad a = r\left[\frac{2 - 1 + \frac{l}{k}}{1 - \frac{l}{k}}\right] = r\frac{\left(1 + \frac{l}{k}\right)}{\left(1 - l/k\right)}$$

$$\text{Reflection Probability} \equiv \left|\frac{r}{a}\right|^2 = \frac{(1 - l/k)^2}{(1 + l/k)^2}$$

but $$\frac{l}{k} = \sqrt{\frac{8 m V_0}{3 \hbar^2}}\sqrt{\frac{3 k^2}{2 m V_0}} = 2$$

thus $\rightarrow$ Reflection Probability $\equiv \frac{1}{9}$

For a Classical particle there is no reflection.